CE/ARCE 2XX Artificial Intelligence Literacy for Civil and Architectural Engineering (3 Units)

Spring semester · Two 75-minute meetings per week (a lecture day and a lab day) · Location and time: TBD

Description of Course

This course teaches the foundations of artificial intelligence (AI) and how to use AI responsibly in civil and architectural engineering (CE/ARCE) coursework and practice. Students explore the AI landscape, machine learning, neural networks, generative AI, and AI agents, and examine their applications, tools, responsible use, ethics, and policy in CE/ARCE. Every week pairs a lecture with a hands-on computer lab in Python (Google Colab) using real engineering data — concrete mixes, building energy, bridges, traffic, and construction-safety rules. Students learn to understand core AI concepts, run and interpret AI models, verify AI outputs, and recognize their limitations.

Course Prerequisites

None. No prior programming experience is required; the labs start from the very first line of Python.

Instructor and Contact Information

Dr. Liang Zhang, Civil Engineering 220F, (520) 621-8522, liangzhang1@arizona.edu

Office Hours: TBD (including at least one evening or online slot), and by appointment

Course homepage (all course materials): https://github.com/TensorBuildLab/AI-Literacy-CE-ARCE

Course Format and Teaching Methods

The class meets twice a week for 75 minutes. **Lecture day:** an interactive lecture with CE/ARCE examples, a short concept quiz on the previous week, and in-class activities. **Lab day:** a guided computer lab in Google Colab (bring a laptop) that ends with an **exit ticket** — three short questions you answer alone, without AI, in the last 10 minutes of lab. Other methods: individual homework, a team project, class discussions, and web-delivered course content.

Course Objectives

During this course, students will:

- Attend lectures and discussions to learn the fundamental concepts of artificial intelligence, machine learning, neural networks, generative AI, and AI agents, and their applications in civil and architectural engineering.
- Perform hands-on computer lab exercises that progress from basic Python programming and working with data to running machine learning, neural network, generative AI, and agent models and understanding how they work.
- Reason about the responsible use, verification, ethics, and policy of AI in civil and architectural engineering coursework and practice.
- Engage in project-based learning to apply AI to civil and architectural engineering problems.

Expected Learning Outcomes (Undergraduate)

Course learning outcomes (CLOs) and the ABET student outcomes they support:

CLO	By the end of the course, students will be able to…	ABET	Where
1	Explain what AI is and is not, and place machine learning (ML), deep learning (DL), and generative AI (GenAI) in the AI landscape using precise vocabulary.	7	Wk 1–2
2	Explain how a machine learns from data (model, cost function, gradient descent, train/test evaluation) and apply it to a civil or architectural engineering dataset in Python.	1, 6	Wk 3–4
3	Describe how neural networks, including convolutional and sequence models, make predictions and are trained, and run and modify them on engineering data.	1, 6	Wk 5–6
4	Explain at a conceptual level how generative AI works, why it can be	1, 7	Wk 7–8

	wrong, and use prompting and retrieval to get better results.		
5	Describe the parts of an AI agent, build a simple one, and judge when an agent is appropriate.	1, 7	Wk 10
6	Identify AI applications across CE/ARCE sub-disciplines, and select and critically evaluate AI tools for an engineering task.	7	Wk 11–12
7	Verify AI outputs, recognize limitations (data shift, bias, uncertainty), and apply responsible-use practices, professional ethics, and policy.	4, 6	Wk 8, 12–14
8	Work in a team to apply AI to a CE/ARCE problem and communicate the approach, results, and limitations to technical and non-technical audiences.	3, 5	Project

The Accreditation Board for Engineering and Technology (ABET) accredits the Civil Engineering curriculum at the University of Arizona. This course fits in the Civil Engineering curriculum and supports the following ABET outcomes:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
3. An ability to communicate effectively with a range of audiences
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies

Absence and Class Participation Policy

The UA's policy concerning Class Attendance and Participation is available at: https://catalog.arizona.edu/policy/courses-credit/courses/class-attendance-participation

The UA policy regarding absences for any sincerely held religious belief, observance or practice will be accommodated where reasonable: https://policy.arizona.edu/employment-human-resources/religious-accommodation-policy.

Absences pre-approved by the UA Dean of Students (or Dean Designee) will be honored. See: https://deanofstudents.arizona.edu/support/student-assistance

Participating in the course and attending lectures and labs are vital to the learning process. As such, attendance is required at all lecture and lab meetings. Absences may affect a student's final course grade. If you anticipate being absent, are unexpectedly absent, or are unable to participate in class online activities, please contact me as soon as possible. To request a disability-related accommodation to this attendance policy, please contact the Disability Resource Center at (520) 621-3268 or drc-info@arizona.edu. If you are experiencing unexpected barriers to your success in your courses, the Dean of Students Office is a central support resource for all students and may be helpful. The Dean of Students Office is located in the Robert L. Nugent Building, room 100, or call 520-621-7057.

Course Communications

Course materials — lecture slides, lab notebooks, homework handouts, the project guide, and datasets — are posted on the course GitHub page: https://github.com/TensorBuildLab/AI-Literacy-CE-ARCE. Brightspace (D2L, https://d2l.arizona.edu) is used for announcements, quizzes, submissions, and grades. It is the student's responsibility to check both regularly. Please use your UA email and include "CE/ARCE 2XX" in the subject line. For course questions others may share, use the Brightspace discussion board.

Required Texts or Readings

There is no required textbook. Short, free readings and videos are listed for each week on the course GitHub page (for example, the free online course Elements of AI, https://www.elementsofai.com). Readings are not graded directly, but they prepare you for the quizzes, labs, and class discussions.

Required Materials

- **A laptop for every lab day** (Windows, Mac, or Chromebook with a current web browser). A tablet is not enough. If this is a barrier, talk to the instructor; the UA Libraries also lend laptops: https://lib.arizona.edu/borrow/tech.
- **Required software: Google Colab**, a cloud-based Python notebook environment. No local installation is required; a free Google account is needed. Free access at https://colab.research.google.com.
- **Optional software: Anaconda**, for students who prefer to run notebooks locally. Free download at https://www.anaconda.com.
- No paid AI subscriptions are required. Labs use free, open models that run inside Colab on Google's computers, so model downloads do not use your laptop storage or phone data plan.

Assignments: Schedule/Due Dates

All assignments are shown in the course schedule below and on Brightspace. Unless stated otherwise, lab notebooks are due Sunday at 11:59 pm (Arizona time) at the end of the lab's week, and homework is due Sunday at 11:59 pm at the end of the week shown.

Late work. Each student has two **grace days** per semester, usable on any lab or homework (both on the same item is fine). Each grace day extends a deadline by 24 hours with no explanation needed; write "grace day used" in the Brightspace submission comment. Late days are counted from the original deadline. Exceptions are made for documented emergencies, religious observances, and Disability Resource Center accommodations — contact the instructor as early as possible.

Item	Deadline	Grace days?	After the deadline
Lab notebook (including exit ticket)	Sunday 11:59 pm of the lab's week	Yes	−10% per day; not accepted after 3 days
Homework	Sunday 11:59 pm of the week shown	Yes	−10% per day; not accepted after 3 days
Project files (proposal + charter, check-in form, slides, final files)	As in the schedule	No	−10% per day; not accepted after 3 days
In-class events (quizzes, midterm, Week 12–13 check-in, Week 15 dry run)	In class	No	Excused: make-up within one week. Otherwise the points are lost (the two lowest quizzes are dropped).

Midterm Exam

The midterm is a closed-book, in-class exam held during the Week 7 lab session (75 minutes). It covers Weeks 1–6: the AI landscape, machine learning, and neural networks. It emphasizes conceptual understanding and short calculations over memorization; a formula sheet is provided and a basic calculator is allowed (any non-graphing, non-programmable calculator; no phones, laptops, or AI tools). A study guide and practice exam are posted on the course GitHub page. Make-up exams are given only for documented reasons, arranged in advance when possible.

Final Project

Teams of 3–4 students apply AI to a civil or architectural engineering problem in one of four tracks: **Predict** (machine learning on engineering data), **See** (computer vision), **Assist** (a generative-AI or agent tool for an engineering workflow, with evaluation), or **Evaluate** (a systematic test of an AI tool on an engineering task). Every project includes a baseline, an

evaluation, verification of results, a responsible-AI analysis, and an AI-use statement.

Milestones: team registration (Week 10), proposal and team charter (Week 11, 10% of the project grade), progress check-in (Week 12 or 13 lab day, in two waves, 10%), dry run with peer feedback (Week 15, 5%), final presentation (30%), final notebook and report of at most 6 pages (35%), and individual reflection and peer evaluation (10%). Individual grades may be adjusted based on peer evaluation, using the rule published in the project guide.

The final presentations take place during the final exam period: about 10 minutes per team plus questions. With more than eight teams, presentations run in two parallel rooms or as 8-minute talks; the schedule is posted by Week 13. The Final Exam Regulations and Final Exam Schedule can be found at https://registrar.arizona.edu/finals.

Grading Scale and Policies

University policy regarding grades and grading systems is available at https://catalog.arizona.edu/grading-policies.

The instructor will assess the student outcomes by grading the following coursework, to be completed by all students:

Component	Weight
Computer Labs (12; lowest dropped)	25%
Homework (5; lowest dropped)	20%
Midterm Exam	20%
Final Project	25%
Lecture Participation & Quizzes (lowest two quizzes dropped)	10%
TOTAL	100%

Within each category, assignments are weighted equally, although it is acknowledged they may be of varying difficulty. Students will evaluate the course and instructor at the end of the semester to provide feedback on the perceived quality of the course and the effectiveness of the instructor. Grades will be assigned as follows, although a curve may be applied: A = 90% and above; B = 80% to 89.99%; C = 70% to 79.99%; D = 60% to 69.99%; E = below 60%. Please note that there is a very clear cut between grades. For example, 89.99% is a B.

Computer Labs: the 12 graded lab sessions (Labs 01–06, 08, and 10–14) use Python and Google Colab; Lab 15 (project dry runs) counts toward the final project. Each lab is evaluated on completion — finishing the notebook's exercises and submitting it — and on the end-of-class exit ticket. The lowest lab grade is dropped.

Homework: five homework assignments are completed and submitted individually. Each person must turn in their own work, including any code, output, and written responses. Each assignment shows an AI-use label for each part. The lowest homework grade is not included in the final grade calculation. Extra-credit opportunities may arise throughout the semester.

Lecture Participation & Quizzes: short concept quizzes (5–8 minutes) at the start of most lecture days cover the previous week's material; in-class activities and discussions count toward participation. The lowest two quiz grades are dropped. Preparation takes the form of short readings or videos; these are not directly evaluated but are necessary for the labs, quizzes, and class discussions.

Midterm Exam: see Midterm Exam above.

Final Project: see Final Project above. Rubrics are in the project guide on GitHub.

Requests for incomplete (I) or withdrawal (W) must be made in accordance with University policies, available at https://catalog.arizona.edu/grading-policies.

Dispute of Grade Policy: Grade disputes on homework, labs, exams, or the project should be brought to the attention of the course instructor no later than 7 days after a grade is received, either online or written in person.

Schedule

Dates will be added when the semester calendar is set. Topics may shift slightly; changes are announced on Brightspace.

Wk	Lecture day	Lab day	Due / notes (Sunday 11:59 pm unless noted)
1	What AI Is (and Isn't): course introduction, definitions, the big picture	Lab 01 — Python On-Ramp: Colab, Python basics; run a vision model and a language model	Lab 01; pre-course survey
2	Zooming In: Machine Learning, Deep Learning, and Generative AI	Lab 02 — Working with Data: load, explore, and visualize concrete-mix data	Quiz on Wk 1; Lab 02; HW1 posted
3	How Machines Learn 1: Linear Regression and the Cost Function	Lab 03 — Model Representation and Cost	Quiz on Wk 2; Lab 03; HW1 due
4	How Machines Learn 2: Gradient Descent and Generalization	Lab 04 — Gradient Descent: from scratch and with scikit-learn	Quiz on Wk 3; Lab 04; HW2 posted
5	Neural Networks 1: Neurons, Layers, and Training	Lab 05 — Your First Neural Network (building energy)	Quiz on Wk 4; Lab 05; HW2 due
6	Neural Networks 2: Seeing and Remembering — CNNs and Sequence Models	Lab 06 — Crack Detection with a CNN; traffic forecasting	Quiz on Wk 5; Lab 06; HW3 posted
7	Generative AI 1: How Large Language Models Work	MIDTERM EXAM (Weeks 1–6), in the lab session	Quiz on Wk 6 (lecture day); midterm (lab day)
8	Generative AI 2: Prompting, Retrieval (RAG), and Images	Lab 08 — Inside an LLM, and RAG over OSHA excavation rules	Quiz on Wk 7; Lab 08; HW3 due; HW4 posted; mid-semester survey
9	Spring Break — no class	Spring Break — no class	—
10	AI Agents: From Answering to Acting	Lab 10 — Build a Simple Engineering Agent	Quiz on Wk 8; Lab 10; HW4 due; project assigned; team registration due
11	AI Across Civil and Architectural Engineering: The Big Picture	Lab 11 — End-to-End Machine Learning on Arizona Bridges (project template)	Quiz on Wk 10; Lab 11; project proposal + team charter due
12	AI Tools for Engineers: Choosing, Using, and Verifying	Lab 12 — Trust but Verify; project check-ins (wave 1)	Quiz on Wk 11; Lab 12; HW5 posted; check-in form due before your check-in
13	Responsible AI: Reliability, Bias, and Transparency	Lab 13 — Stress-Testing Models; project check-ins (wave 2)	Quiz on Wk 12; Lab 13
14	AI Policy, Ethics, and Professional Practice	Lab 14 — AI Policy Workshop; project work session	Quiz on Wk 13; Lab 14; HW5 due
15	Synthesis: The Big Picture, Revisited	Lab 15 — Project Dry Runs (5-minute practice talks) and Peer Feedback	Quiz on Wk 14; dry-run slides due before lab; course evaluations
16	Final Project Presentations (final exam period)	—	Final notebook, report, AI-use statement, reflection, peer evaluation

Use of Generative AI: Sometimes Permitted (by Assignment Label)

This course is about AI, so using AI is part of the learning — under clear rules. Every assignment carries one of three labels:

- **AI-Free** — do not use AI tools to produce what you submit. Applies to quizzes, the midterm exam, lab exit tickets, and homework parts labeled AI-Free. You may still use AI to study the concepts beforehand.
- **AI-Assisted** — you may use AI tools for help such as explanations, debugging, and brainstorming. Include an AI-use acknowledgment; you are responsible for every line you submit. This is the default for labs, most homework parts, and the final project.
- **AI-Required** — using an AI tool is part of the task (for example, testing a language model's answers). Follow the instructions and document your prompts and the outputs.

In all cases, you may NOT: submit AI-generated work as your own without acknowledgment; use AI to generate answers in place of participating in class discussions, debates, or activities; let AI write, analyze, or interpret the final project to the point that the work no longer reflects your team's own understanding; or paste other people's private information, exam content, or confidential data into AI tools.

How to write an acknowledgment statement: "I acknowledge the use of [name of AI system(s)] to [describe the specific use]. Prompts used included [share a link or copy-and-paste the prompts]. I checked the output by [how I verified it]. I am responsible for this work." See the UA Libraries guide Citing generative AI: https://libguides.library.arizona.edu/students-chatgpt/citing.

If you are unsure whether a use is allowed, ask before you submit. Undisclosed or prohibited AI use is a violation of the Code of Academic Integrity.

Classroom Behavior Policy

To foster a positive learning environment, students and instructors have a shared responsibility. We want a safe, welcoming, and inclusive environment where all of us feel comfortable with each other and where we can challenge ourselves to succeed. To that end, our focus is on the tasks at hand and not on extraneous activities (e.g., texting, chatting, making phone calls, web surfing, etc.). Laptops are required on lab days; on lecture days, please use them only for course activities.

Students are asked to refrain from disruptive conversations with people sitting around them during lectures. Students observed engaging in disruptive activities will be asked to cease this behavior. Those who continue to disrupt the class will be asked to leave the lecture or lab and may be reported to the Dean of Students.

Threatening Behavior Policy

The UA Threatening Behavior by Students Policy prohibits threats of physical harm to any member of the University community, including to oneself. See https://policy.arizona.edu/education-and-student-affairs/threatening-behavior-students.

Accessibility and Accommodations

At the University of Arizona, we strive to make learning experiences as accessible as possible. If you anticipate or experience barriers based on disability or pregnancy, please contact the Disability Resource Center (520-621-3268, https://drc.arizona.edu/) to establish reasonable accommodations. Course slides are provided as PowerPoint and PDF files and lab notebooks as text-based notebooks; contact the instructor if you need another format.

Code of Academic Integrity

Students are encouraged to share intellectual views and discuss freely the principles and applications of course materials. However, graded work/exercises must be the product of independent effort unless otherwise instructed. Students are expected to adhere to the UA Code of Academic Integrity: https://deanofstudents.arizona.edu/policies/code-academic-integrity.

The University Libraries have tips for avoiding plagiarism: https://lib.arizona.edu/research/sources/plagiarism.

Selling class notes and/or other course materials to other students or to a third party for resale is not permitted without the instructor's express written consent. Violations to this and other course

rules are subject to the Code of Academic Integrity and may result in course sanctions. Additionally, students who use D2L or UA e-mail to sell or buy these copyrighted materials are subject to Code of Conduct Violations for misuse of student e-mail addresses. This conduct may also constitute copyright infringement.

UA Nondiscrimination and Anti-harassment Policy

The University is committed to creating and maintaining an environment free of discrimination; see https://policy.arizona.edu/employment-human-resources/nondiscrimination-and-anti-harassment-policy.

Our classroom is a place where everyone is encouraged to express well-formed opinions and their reasons for those opinions. We also want to create a tolerant and open environment where such opinions can be expressed without resorting to bullying or discrimination against others.

Additional Resources for Students

- UA academic policies and procedures: https://catalog.arizona.edu/policy/all
- UA syllabus policies: https://catalog.arizona.edu/syllabus-policies
- Student Assistance (Dean of Students): https://deanofstudents.arizona.edu/support/student-assistance
- Counseling & Psych Services: https://health.arizona.edu/counseling-psych-services
- Campus Pantry: https://campuspantry.arizona.edu/

Confidentiality of Student Records

https://registrar.arizona.edu/privacy-ferpa/ferpa

Subject to Change Statement

Information contained in the course syllabus, other than the grade and absence policy, may be subject to change with advance notice, as deemed appropriate by the instructor.